

The Zeta Circuit

(s) as a composite of resistances, and the two companion notes as its first two logarithmic derivatives

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Status. Companion note under construction, section by section. This is the opening scope and the foundational verification (§§1–2) only; a later section pursues the one open question — whether the primon cumulants carry arithmetic structure beyond generic moment theory — under criteria committed in advance. Nothing here is promoted to the project spine.

1. Scope and boundaries

What this note does. The companion notes read discrete operations through the two-legged ledger and price their erasures. This note turns the same reading on the Riemann zeta function — not as an object to be interrogated for its zeros, but as a **circuit** assembled from the framework’s primitives: prime resistances, a composition rule, and the ledger. The organizing claim, verified in §2 and developed after: **(s) is the partition function of a gas of erasure-priced integers, and the two companion notes are its first two logarithmic derivatives** — Note A (innovation) at the first, Note B (fluctuation) at the second. [Reading]

The primon reading (known physics, with the framework’s one turn). Via the Euler product $(s) = \sum_n n^{-s} = \prod_p (1 - p^{-s})^{-1}$, the statistical theory of numbers treats each integer n as a state of energy $\ln n$, each prime p as a fundamental mode of energy $\ln p$, and s as an inverse temperature, with (s) its partition function (Julia; Spector; and, in its operator form, the Bost–Connes system). This much is established. [Fact, known] The framework adds one identification: by the resistance isomorphism $\Omega(1/n) = \ln n$, the energy $\ln n$ is the Landauer erasure cost of the integer. So the primon gas is a gas of integers weighted by their erasure cost, and $(s) = \sum_n e^{-s \cdot (\text{erasure cost of } n)}$. [Reading]

Regime framing. The per-integer erasure cost $\ln n$ is a Regime-1 quantity — a discrete, quantized cost carried by an individual integer. Every *cumulant* computed in this note is a Regime-2 quantity: an average over the primon Gibbs ensemble at temperature s , a probe-side statistic, never a property of the genesis layer. The banked one-liner holds — the ensemble is probe-manufactured; the integers’ own costs are the quantized ledger.

Standard identities are not the content. The relations used here — that the cumulants of the energy are the logarithmic derivatives of the partition function — are textbook statistical mechanics, exact and standard. As in the companion notes, we say this plainly so it is never mistaken for a discovery. The content is the **erasure reading**: that the first log-derivative is an innovation spectrum, the second a fluctuation, and that the circuit’s steps sort into reversible and dissipative on the ledger.

Boundaries.

1. **No leverage on the Riemann Hypothesis.** This is the load-bearing boundary, stated as firmly as the corpus states it elsewhere. Nothing here bears on the location of ζ 's zeros or the critical line. The dissection lives entirely in the **convergent regime** $s > 1$, in the analytic structure of the logarithmic derivatives; it reads ζ 's thermodynamic and erasure structure, not its zeros. The Weyl-clock / phasor-bouquet correspondence already in the corpus carries an explicit no-leverage flag, and that flag stays up here.
2. **The physics is borrowed; the reading is the contribution.** The primon gas is an established model; this note does not claim it. What it claims is the erasure interpretation of the model's quantities and their assembly into a circuit.
3. **Cartography, not evidence.** As throughout, the circuit is a reading; that ζ hosts an erasure interpretation says nothing about whether the genesis system holds beyond the axioms that posit it.

Grading. Every claim carries an inline grade — **[Fact]** (a verified identity or established result), **[Reading]** (a framework-consistent interpretation), **[Lead]** (an open thread with a kill-or-promote criterion). §2's identities are Facts, verified against high-precision ground truth; the erasure readings are Readings; the one genuinely open question — arithmetic structure in the cumulants — is carried as a Lead and resolved, under pre-registered criteria, in a later section.

2. The primon circuit and its first two cumulants

The circuit. The components are the framework's own. Each prime p is a resistance of value $\ln p$. The composition rule is the Euler product: the primes combine multiplicatively, $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$, which is the same object as the Dirichlet sum $\sum_n n^{-s}$ by unique factorization. The reading is $\zeta(s) = \sum_n e^{-s \cdot \ln n}$ — the partition function over integer erasure costs.

The reversible seam. **[Fact $\rightarrow$ Reading]** The Euler product and the Dirichlet sum are equal because every integer factors uniquely into primes. We verified this directly: the product over primes $p \leq P$ equals the sum over P -smooth integers (those with no prime factor above P), agreeing to 10^{-11} across $P = 2, 3, 5, 7, 13$, both climbing toward $\zeta(s)$ as P grows. **[Fact]** On the ledger this seam is *reversible*: factoring and re-multiplying is a lossless re-encoding of the same number, and reversible operations are free. The circuit's compositional wiring costs nothing; the price, when it comes, is elsewhere. **[Reading]**

The first cumulant: mean erasure cost is the innovation spectrum. **[Fact]** Under the Gibbs measure $P_s(n) = n^{-s} / \zeta(s)$, the mean energy is the mean erasure cost, and it is the first logarithmic derivative of ζ :

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_n \ln n \cdot P_s(n) = -\frac{\zeta'(s)}{\zeta(s)} = \sum_n \Lambda(n) P_s(n),$$

the von Mangoldt Dirichlet series. We computed all three expressions independently at $s = 2$ and $s = 3$: the Gibbs mean, the von Mangoldt series, and a high-precision evaluation of $-\zeta'/\zeta$ agree — to ten–twelve digits at $s = 3$ (where the truncation converges fast), to five–seven at $s = 2$. **[Fact]** The reading writes itself: the von Mangoldt function is exactly Note A's channel-innovation spectrum, so **the mean energy of the primon gas is the innovation spectrum**, and taking the first derivative of ζ is the irreversible act that extracts it — Λ is where the ledger is finally charged. **[Reading]**

The second cumulant: fluctuation, Fisher, heat. [Fact] The variance of the erasure cost is the second logarithmic derivative:

$$= \text{Var}(\ln n) = (\log \zeta)'(s) = 1/s - (\zeta'/\zeta)^2,$$

which is at once the primon gas's heat capacity and the **Fisher information** the ensemble carries about the temperature s — the score of the Gibbs family in s is $-(\ln n - \ln \zeta)$, whose variance is $1/s$. We confirmed the primon variance against $(\log \zeta)'$ at both temperatures: matching to ten digits at $s = 3$, to five at $s = 2$. [Fact] This is Note B's fluctuation probe made analytic: where the mean ζ' fixes the innovation spectrum, the variance ζ'' is the second-cumulant handle Note B built by hand, here delivered as a concrete number-theoretic function. [Reading]

The ladder, and the two notes as two derivatives. The pattern continues: $\zeta^{(j)} = (-1)^j (\log \zeta)^{(j)}$ is the whole cumulant ladder of Note B §6, now indexed by derivatives of ζ . So the two companion notes attach to one object at successive derivatives — Note A at the first log-derivative (innovation, von Mangoldt), Note B at the second and beyond (fluctuation, the ladder). The circuit is: prime resistances $\ln p \rightarrow$ Euler-product composition (reversible) $\rightarrow$ differentiate once for the innovation spectrum, again for the fluctuation. [Reading] What remains open — taken up next under committed criteria — is whether these primon cumulants carry structure specific to the primes, beyond the generic moment relations any energy spectrum would satisfy. [Lead — open]

3. Independent prime modes, and where the arithmetic lives

Z1 established that the primon cumulants are the logarithmic derivatives of ζ . The question §1 flagged as open: do those cumulants carry structure *specific to the primes*, beyond the generic moment relations any energy spectrum satisfies? This section answers it — for the convergent regime — and the answer sharpens the picture rather than promoting it.

The primon gas is a system of independent prime modes. [Fact] Because $\log \zeta(s) = \sum_p -\log(1 - p^{-s})$, the gas factors into one independent mode per prime: prime p is a geometric (bosonic) oscillator of quantum energy $\ln p$, occupied n_p times with probability $(p^{-s})^{n_p}$, and the total erasure cost is $E = \sum_p n_p \ln p$. Every cumulant is therefore additive over primes,

$$\zeta^{(j)}(s) = \sum_p (\ln p)^j \cdot g_j(p^{-s}),$$

with g_j the j -th cumulant of a geometric distribution. We verified this against the ground truth $(-1)^j (\log \zeta)^{(j)}$ at $s = 2$ and $s = 3$, matching to ten–thirteen digits where the truncation converges. [Fact] The first cumulant is the cleanest case: $\zeta' = \sum_p \ln p \cdot p^{-s}/(1 - p^{-s}) = \sum_p \Lambda(n) n^{-s}$, so the von Mangoldt series is exactly the summed mean occupation energy of the prime modes. This is the arithmetic realisation of the picture Note A §7 and the F1 follow-up circled — the primes as the independent generators of a free commutative monoid, here the independent oscillators of the primon gas. [Reading]

The cumulant-generating function is a shift of $\log \zeta$. [Fact] The same independence gives $K(t) = \log e^{-tE} = \log \zeta(s - t) - \log \zeta(s)$, so the entire ladder is the local Taylor expansion of $\log \zeta$ at s ; the Taylor coefficients reproduce the cumulants to ten digits for $j = 1..4$.

The leading singularity is universal. [Fact] From ζ 's simple pole at $s = 1$ with residue

$1, \dots, j(s) \sim (s-1)^j \rightarrow (j-1)! \text{ as } s \rightarrow 1$ — verified at $s = 1.1, 1.01, 1.001$ (0.999, 1.000, 2.0, 6.0 for $j = 1\dots4$ at $s = 1.001$). This leading behaviour reflects only that the integers have density 1; it is shared by any independent-mode system whose zeta has a simple pole of residue 1, and is therefore *not* arithmetic-specific.

Where the arithmetic does live: the spectrum and the subleading constants. [Fact] The prime content shows up not in the cumulant *algebra* but in the *values* it is fed. The subleading term of $\zeta(s)$ at the pole is the Euler–Mascheroni constant — $\zeta(s) \sim 1/(s-1) + \gamma$ (verified: -0.57720 against $\gamma = -0.57722$); deeper terms carry prime-zeta constants. These are genuine arithmetic quantities, but they are constants set by the spectrum $\{\ln p\}$, not relations among the cumulants.

The algebra is generic. [Fact] To confirm the cumulant algebra carries no prime-specific relation, we recomputed the standardised cumulants (skewness, excess kurtosis) at $s = 2$ for two null mode systems of the same form — fake primes $q_k \sim k \ln k$ (density-matched) and all integers $m \geq 2$. The primon values (skew 2.344, exkurt 7.627) sit *between* the fake-prime (2.458, 8.349) and integer (2.015, 5.374) values: all three are generic geometric-mode shapes, and nothing singles the primes out. [Fact]

Verdict — parked, and §6’s park extended. No cumulant-level relation distinguishes the primon gas from a generic independent-geometric-mode gas: the mode decomposition, the CGF-as-shift, the universal pole, and the standardised-cumulant character are all matched by the nulls. The arithmetic lives entirely in the spectrum $\{\ln p\}$ and in subleading analytic constants (γ , prime-zeta values), never in the ladder algebra. Note B §6 parked the cumulant ladder as classical moment theory on content-free cascades; Z2 shows that carrying *real primes* in the cumulants does not change the verdict — the algebra is generic even here. [Lead — parked]

The one place not looked. [boundary] There is exactly one place where non-generic prime structure genuinely resides: the *analytic continuation*. The function $\zeta(s) = \sum n^{-s}$, continued past $s = 1$, has poles at the nontrivial zeros of ζ — this is the content of the classical explicit formula, and it is where the primes’ fine structure lives. That structure is real, and it is entirely classical (Riemann; von Mangoldt). It is also precisely the critical-strip territory this note’s boundary places out of scope: the location of those zeros is the Riemann Hypothesis, on which the framework offers descriptive correspondence (the corpus’s Weyl-clock / phasor-bouquet bridge) and no leverage. We name the structure to be honest about where it lives, and we do not cross into it here. [Fact, classical] that the continuation’s poles are the zeros; [boundary] that this yields no purchase on their location.

Appendix — verbatim script outputs

Built incrementally, one block per experimental step; every number in the prose above appears here.

Z1 · z1_primon_circuit.py (self-authored, PID-named)

Three independent routes to $\zeta(s)$ (Gibbs mean, von Mangoldt series, $\text{mpmath} - /$) agree; ζ matches $\text{mpmath}(\log)$; the Euler product equals the P-smooth sum.

truncation N = 2000000; mpmath ground truth: True

--- Cumulants of the primon energy ($E = \ln n$) under Gibbs measure n^{-s}/ζ ---

```

s = 2.0
zeta_N(s) = 1.6449335668
kappa_1 mean energy (-zeta'/zeta) = 0.5699564523
kappa_1 via sum Lambda(n) n^-s = 0.5699604931 |diff| = 4.04e-06
kappa_2 variance = Fisher = C_heat = 0.8844139649
kappa_3 = 1.9491208924
mpmath truth: -zeta'/zeta = 0.5699609931, (log zeta)'' = 0.884481834
|mean - truth| = 4.54e-6; |var - truth| = 6.79e-5

```

```

s = 3.0
zeta_N(s) = 1.2020569032
kappa_1 mean energy (-zeta'/zeta) = 0.1648226822
kappa_1 via sum Lambda(n) n^-s = 0.1648226822 |diff| = 3.14e-12
kappa_2 variance = Fisher = C_heat = 0.1722807115
kappa_3 = 0.2215047317
mpmath truth: -zeta'/zeta = 0.1648226822, (log zeta)'' = 0.1722807115
|mean - truth| = 3.54e-12; |var - truth| = 2.13e-11

```

```

--- Reversible re-encoding: Euler product over p<=P vs sum over P-smooth n<=N ---
P= 2: Euler product = 1.3333333333 P-smooth sum = 1.3333333333 |diff| = 1.94e-11
P= 3: Euler product = 1.5000000000 P-smooth sum = 1.4999999998 |diff| = 2.14e-10
P= 5: Euler product = 1.5625000000 P-smooth sum = 1.5624999990 |diff| = 1.02e-09
P= 7: Euler product = 1.5950520833 P-smooth sum = 1.5950520802 |diff| = 3.13e-09
P=13: Euler product = 1.6179176613 P-smooth sum = 1.6179176490 |diff| = 1.23e-08

```

Note: an earlier run used `mp.zeta(s, 1)` for the cross-check — the Hurwitz zeta, not the derivative — which printed an absurd $- / = -1.0$; the script was corrected to `mp.diff`, and the corrected ground truth (shown above) confirms the primon computation, which was independently self-consistent throughout.

Z2 · z2_primon_arithmetic.py (self-authored, PID-named; parked Lead)

Decomposition matches ground truth; CGF = shift of $\log$; universal pole $(j-1)!/(s-1)^j$; sub-leading constant - ; standardised cumulants generic across primon / fake-prime / integer mode systems.

```

(1) Independent-prime-mode decomposition kappa_j = sum_p (ln p)^j g_j(p^-s)
s=2.0
kappa_1: mode-sum=0.5699604932 truth=0.5699609931 |diff|=5.00e-07
kappa_2: mode-sum=0.8844740819 truth=0.8844818340 |diff|=7.75e-06
kappa_3: mode-sum=1.9500151080 truth=1.9501358327 |diff|=1.21e-04
kappa_4: mode-sum=5.9668314706 truth=5.9687202695 |diff|=1.89e-03
s=3.0
kappa_1: mode-sum=0.1648226822 truth=0.1648226822 |diff|=4.02e-13
kappa_2: mode-sum=0.1722807115 truth=0.1722807115 |diff|=1.93e-12
kappa_3: mode-sum=0.2215047320 truth=0.2215047321 |diff|=2.82e-11
kappa_4: mode-sum=0.3606578108 truth=0.3606578112 |diff|=4.24e-10

```

(2) CGF = shift: $\kappa_j = j! \cdot [t^j](\log \zeta(s-t) - \log \zeta(s))$ at $s=2$
 κ_1 via CGF = 0.5699609931 truth = 0.5699609931
 κ_2 via CGF = 0.884481834 truth = 0.884481834
 κ_3 via CGF = 1.950135833 truth = 1.950135833
 κ_4 via CGF = 5.968720269 truth = 5.968720269

(3) Universal pole: $\kappa_j(s) \cdot (s-1)^j \rightarrow (j-1)!$ as $s \rightarrow 1+$
 $s=1.1$: $j=1:0.94410364(\text{target } 1)$ $j=2:0.99822366(\text{target } 1)$ $j=3:1.999905(\text{target } 2)$
 $s=1.01$: $j=1:0.99424655(\text{target } 1)$ $j=2:0.99998135(\text{target } 1)$ $j=3:1.999999(\text{target } 2)$
 $s=1.001$: $j=1:0.99942297(\text{target } 1)$ $j=2:0.99999981(\text{target } 1)$ $j=3:2.0(\text{target } 2)$ $j=4:$

(4a) Subleading constant: $\kappa_1(s) - 1/(s-1) \rightarrow -\gamma$
 $s=1.01$: $\kappa_1 - 1/(s-1) = -0.5753453567$ ($-\gamma = -0.5772156649$)
 $s=1.001$: $\kappa_1 - 1/(s-1) = -0.5770281702$ ($-\gamma = -0.5772156649$)
 $s=1.0001$: $\kappa_1 - 1/(s-1) = -0.5771969097$ ($-\gamma = -0.5772156649$)

(4b) Null comparison - standardized cumulants at $s=2$ ($\text{skew}=k_3/k_2^{1.5}$, $\text{exkurt}=k_4/k_2^2$)
primes (primes) $k_2=0.88447$ $\text{skew}=2.34428$ $\text{exkurt}=7.62735$
fake-primes $k \ln k$ $k_2=0.87622$ $\text{skew}=2.45808$ $\text{exkurt}=8.34879$
all-integers $m \geq 2$ $k_2=2.16096$ $\text{skew}=2.01502$ $\text{exkurt}=5.37412$

End of current draft of The Zeta Circuit. Sections 1–3 complete with a verbatim appendix (Z1, Z2). One Lead parked (§3, the cumulant algebra is generic). The analytic-continuation structure is named and placed out of scope by the no-leverage boundary. Nothing here is promoted to the project spine.